



EXPLAINER NOTE

Why We Request Your System Architecture

Government Systems Prototype Showcase; Explainer Note for Applicants

This note explains why the Ministry of ICT and National Guidance (MoICT&NG) requests a system architecture diagram or description as part of the Government Systems Prototype Showcase application. It is addressed to all applicants and is intended to help you prepare a submission that clearly communicates your system's design.

1. WHAT IS A SYSTEM ARCHITECTURE DIAGRAM

A system architecture diagram is a visual or structured representation of how your system is built and how its components interact. It does not need to be produced by specialist software or drawn to any specific standard. What matters is that it clearly shows:

1. The main components of your system for example, the web or mobile front-end, the back-end application server, the database, and any external services or APIs the system connects to.
2. How those components communicate with each other for example, which components call which, how data flows between them, and what protocols are used.
3. Where data is stored and how it is secured for example, which databases hold which categories of data, and whether encryption is applied.
4. How users access the system for example, through a browser, a mobile app, a USSD interface, or a combination.
5. Any external systems your solution integrates with for example, the National Identification System (NIRA), Uganda Revenue Authority, or other Government platforms.

A simple block diagram, a hand-drawn sketch that has been photographed clearly, or a structured written description of the above components shall all be acceptable. What the Ministry requires is substance, not presentation.

2. WHY THE MINISTRY REQUESTS THIS INFORMATION

2.1 To Assess Fitness for Government Deployment

A system that works well as a prototype may not be designed to scale to tens of thousands or millions of Government users. The architecture diagram allows the Technical Evaluation Panel to assess whether the system has been designed with scalability in mind, whether the database can handle large data volumes, whether the application can be deployed on multiple servers, and whether the design avoids single points of failure that could bring the system down.

Government systems cannot afford extended downtime. Architecture review is a core tool for identifying fragility before deployment, not after.

2.2 To Assess Interoperability with Uganda's Digital Infrastructure

Uganda's Government digital systems are required to work together. The Ministry is building an integrated digital public infrastructure in which systems from different sectors health, education, land, taxation, identity exchange data through common standards and integration platforms.

The architecture diagram allows the panel to assess whether your system has been designed with integration in mind: whether it exposes APIs that other systems can call, whether it uses data standards that are compatible with Government systems, and whether it can connect to platforms such as the National Identification System or the Government integration layer without requiring a complete rebuild.

2.3 To Assess Security Design

Security in a software system is not just a feature it is an architectural property. A system that stores sensitive citizen data without encrypting it at the database level, or that transmits data between components without using secure protocols, may appear functional in a demonstration but carries serious risks in production.

The architecture diagram allows the panel to see where data enters, flows through, and exits the system, and to assess whether security controls have been applied at each stage. This includes how users are authenticated, how access is controlled across different roles, and how audit trails are maintained.

2.4 To Assess Technology Choices

The technologies chosen to build a system have long-term implications for the Government. Systems built on technologies that are no longer actively maintained, that are proprietary to a single vendor, or that require specialist skills that are scarce in Uganda are difficult and expensive to support over time.

The architecture diagram combined with the technology stack declaration in the application allows the panel to assess whether your system has been built on technologies that are appropriate, maintainable, and transferable to a Government technical team.

2.5 To Inform Future Procurement and Integration Planning

Innovators who are enrolled in the National Innovator Registry shall be available for future Government digitisation assignments. When a Ministry or Department is planning a new system, it shall search the Registry for innovators whose existing work is relevant.

The architecture information you provide shall be stored in your Registry profile and used to match your capabilities to future Government requirements. An innovator who has demonstrated a well-designed, interoperable architecture in the health sector may be identified as a strong candidate for a related assignment in social protection or community health information. Your architecture tells us not just what you have built, but what you are capable of building.

3. WHAT TO INCLUDE IN YOUR ARCHITECTURE SUBMISSION

Your architecture submission shall be uploaded to the Government Digital Registry as part of your application. The following elements shall be included where applicable:

Component	What to Show
Frontend or User Interface	How users access the system browser, mobile app, USSD, SMS. Key screens or workflows.
Backend / Application Layer	Where your business logic runs. The framework or language used. How requests are processed.
Database Layer	What database technology is used. What categories of data are stored. Where the database is hosted.
Authentication and Access	How users log in. Whether multi-factor authentication is used. How different user roles are managed.

APIs and Integrations	Any APIs your system exposes or consumes. Any Government systems or third-party services connected.
Hosting and Infrastructure	Where the system is hosted cloud, on-premise, or hybrid. Country of hosting where applicable.
Security Controls	Encryption in transit and at rest. Firewall or network-level protections. Audit logging.
Data Flow	How data moves between components. Where data enters and exits the system boundary.

4. WHAT THE MINISTRY SHALL NOT DO WITH YOUR ARCHITECTURE

1. The architecture diagram shall be treated as confidential and used only for the purposes of evaluation and Registry profiling.
2. It shall not be shared with third parties, published, or used to inform any competing development work.
3. Panel members who review your architecture are bound by a confidentiality undertaking. Your design shall not be replicated or adapted without your consent.
4. Submission of an architecture diagram does not transfer any intellectual property rights to the Ministry or the Government of Uganda.

5. PRACTICAL GUIDANCE FOR FIRST-TIME APPLICANTS

If you have not produced a formal architecture diagram before, do not be discouraged. The following approach shall produce an acceptable submission:

1. Draw a set of boxes representing the main parts of your system the user interface, the server, the database, and any external services.
2. Draw arrows between the boxes to show how they communicate, and label each arrow with what is being exchanged for example, 'user login request', 'patient record', or 'payment confirmation'.
3. Add brief notes to each box identifying the technology used for example, 'React frontend', 'Django REST API', 'PostgreSQL database hosted on AWS RDS'.
4. Add a note indicating where HTTPS or TLS encryption is applied, and where user authentication takes place.
5. Export or photograph the diagram and upload it as a PDF or image file via the GDR portal.

Free tools such as draw.io, Lucidchart (free tier), or even a clearly photographed whiteboard sketch shall be fully acceptable. The quality of your thinking matters more than the quality of your drawing software.

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